

Gayanara E-Commerce

Data Analytics Portfolio Project Roadmap

Phase 1: Data Quality & Foundation (The "Must-Do" First)

Before extracting insights, the data must be validated and cleaned to ensure accuracy.

- **Idea #10: Audit Kualitas Data (Data Audit)**

Identify missing values (nulls) in columns like *phone*, *material*, and *review_text*. Find duplicate customer records (e.g., same email, multiple IDs) and check for orphaned records in *order_items*.

- **Idea #5: Beresin Kategori Produk (Data Cleaning)**

Standardize the messy 14 product categories down to 6 logical categories to ensure accurate grouping.

Phase 2: Ad-Hoc Business Insights (The Bread & Butter)

Answer core operational questions using the cleaned dataset. These make excellent standalone presentation slides.

- **Idea #1: Top Products vs. Dead Stock**

Identify best-selling items to restock and highlight stagnant inventory for potential discounting.

- **Idea #2: Payment Methods & Courier Mix**

Determine the most used payment gateways and logistics partners to strengthen negotiation positions.

- **Idea #3: Revenue by City/Province**

Map revenue contributions and identify high-AOV (Average Order Value) "hidden gem" cities outside the main metro areas.

- **Idea #6: Promo Analysis**

Evaluate if promo codes actually drive higher AOV or if they are just burning capital on guaranteed conversions.

- **Idea #7: Courier Return Rates**

Identify which logistics providers have the highest return/cancellation rates to evaluate fulfillment issues.

Phase 3: Advanced Analytics (The Resume Builders)

Demonstrate business acumen and advanced logic through segmentation and behavior tracking.

- **Idea #8: RFM Segmentation**

Calculate Recency, Frequency, and Monetary scores to segment customers into actionable groups (e.g., Champions, At Risk, Lost).

- **Idea #9: Cohort Retention**

Build a retention heatmap to track customer repeat purchase behavior over subsequent months based on their first purchase date.

Phase 4: Final Deliverables

Synthesize findings into visual and professional formats.

- **Idea #4: Executive Sales Dashboard**

Combine revenue trends, KPIs (AOV, return rates), and product mix into a single interactive view (using Looker Studio or Power BI).

- **Portfolio Case Study & Deck**

Create a 5-8 slide executive presentation and a detailed Markdown write-up for your GitHub Pages portfolio site.